

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in linkedin.com/in/muhammad-haider-zaidi
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, JavaScript, TypeScript, C++, PHP, Golang

Tools/Frameworks: Vue.js, Next.js, Flask, FastAPI, TensorFlow, PyTorch, OpenAI API, Kubernetes, Docker, AWS SageMaker, TensorRT, Airflow, PySpark, Hive, Sqoop, NiFi

Competence: Full-stack Development, Machine Learning, MLOps, Agentic Workflows, AI-Powered Product Development, Data Engineering, AI Infrastructure, LLM Tooling, Real-Time Systems, Monitoring, Cost/Latency Optimization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Developed novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Built dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using AI algorithms, deployed with Docker and FastAPI.
- Developed a real-time tennis foul detection system using TensorRT, reducing compute costs and latency by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Led educational sessions on machine learning and big data at universities to promote AI adoption.
- Developed POCs using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website providing AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and LLM tools/frameworks for enhanced user interaction.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data ingestion.
- Automated data cleaning workflows to streamline client analytics use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Improved model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Designed and implemented a real-time Bitcoin price forecasting system using time-series analysis and machine learning models.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to create the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem.

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Focused on AI-powered product development, agentic workflows, and machine learning applications.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor.
- Specialized in machine learning, data engineering, and full-stack development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python